SOLAR INSTALLATION AND MAINTENANCE MANAGEMENT SYSTEM

A PROJECT REPORT

Submitted by

SELVAGANAPATHY T

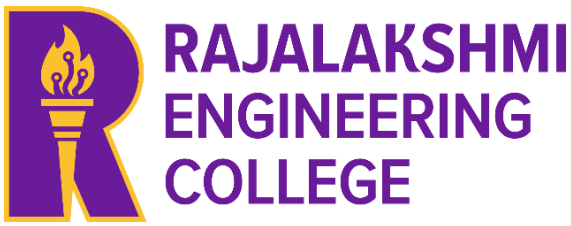
241501194

in partial fulfilment of the End Semester Examination for the course

CB23F35 LOW-CODE APPLICATION DEVELOPMENT USING ZOHO CREATOR

in

BACHELOR OF TECHNOLOGY
ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING



RAJALAKSHMI ENGINEERING COLLEGE
(An Autonomous Institution, Affiliated to Anna University, Chennai)
Rajalakshmi Nagar, Thandalam, Tamil Nadu – 602 105

OCTOBER - 2026

RAJALAKSHMI ENGINEERING COLLEGE
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BONAFIDE CERTIFICATE

Certified that this project report titled **SOLAR INSTALLATION AND MAINTENANCE MANAGEMENT SYSTEM** is the bonafide work of **SELVAGANAPATHY T - 241501194** , who carried out the individual project work on the Zoho Creator low-code platform for the course CB23F35 – Low-Code Application Development using Zoho Creator, during the academic year **2026-2027** Semester **V**, under my supervision.

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INTERNAL EXAMINER

EXTERNAL EXAMINER

DECLARATION

I , **SELVAGANAPATHY T - 241501194** , hereby declare that the project report titled **Solar Installation And Maintenance Management System**, submitted for the End Semester Examination of the course CB23F35 – Low-Code Application Development using Zoho Creator, is a record of original work carried out by me individually. The Zoho Creator application described in this report—including its data model, forms, reports, pages, Deluge scripts and workflows—was designed and developed by me.

I further declare that this work has not been submitted, in part or in full, for the award of any other degree or course credit. Any platform documentation or external literature used in preparing this report is acknowledged in the references. This report distinguishes features evidenced by the supplied application export from activities that still require live verification.

Place: CHENNAI
Date: [DD Month YYYY]

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SELVAGANAPATHY T

ABSTRACT

Solar installation businesses coordinate customer information, system installation records, equipment inventory, technicians, service requests and preventive maintenance. When these activities are tracked in separate spreadsheets or informal messages, teams can experience duplicated records, limited visibility into installation status, delayed service response and difficulty identifying stock constraints.

The Solar Installation And Maintenance Management System addresses this operational coordination problem through a low-code application designed in Zoho Creator. The supplied application export defines seven primary forms: Customer, Solar Installation, Equipment Inventory, Technician, Maintenance Request, Maintenance Schedule and Maintenance Checklist.

These modules provide structured fields for customer contact details, system capacity, panel and inverter information, warranty dates, inventory quantities, technician availability, service priority, diagnosis, action taken and checklist observations. Related reports include standard list reports and grouped views by issue category, installation status, maintenance type, technician specialization, equipment category, customer type and checklist item.

A dashboard presents summary indicators such as average system capacity, maintenance-request count, installation count and maintenance-schedule count, alongside charts and quick links for creating records. The configured Deluge workflows include a validation rule that blocks inventory submission when quantity available is below the minimum stock level, an automation that attempts to assign an available technician to a Critical maintenance request, and email notifications when a customer or technician record is created.

These are evidenced by the exported configuration; successful execution in a live environment must be confirmed during demonstration. The application was reviewed from its text export for module coverage, workflow intent and report/dashboard configuration. This source review is not a substitute for end-to-end execution, role-permission verification or production deployment evidence. Accordingly, measured reductions in processing time, real-user acceptance, uptime and production performance are not claimed in this report.

The proposed test plan includes normal, boundary and negative scenarios for record creation, stock validation, technician assignment, email notification and report

consistency. Before submission, the student should execute these tests in Zoho Creator, record actual outcomes, capture genuine screenshots, verify permission settings, and replace all identity and institution placeholders. The project demonstrates how a low-code platform can combine structured data capture, visual reporting and targeted Deluge scripting for solar installation and maintenance operations.

Keywords: Low-Code Development; Zoho Creator; Deluge Scripting; Solar Installation; Maintenance Workflow; Inventory Validation.

PROJECT AT A GLANCE

Attribute	Details
Project Title	Solar Installation Management System
Application Domain	Solar energy installation, asset records and maintenance service management
Student Name & Register no	
Platform	Zoho Creator (application export supplied as .ds file)
Primary modules	Customer; Solar Installation; Equipment Inventory; Technician; Maintenance Request; Maintenance Schedule; Maintenance Checklist
Form count	7 primary forms evidenced in the export
Report coverage	Standard reports and grouped/kanban reports across the seven forms
Dashboard	DashBoard page with KPI tiles, quick-add links, scatter/bar charts and maintenance request report
Automation	Critical-priority technician assignment; low-stock validation; customer email; technician email
Scripting language	Deluge
External integrations	No external REST/CRM integration is evidenced in the supplied export
Deployment status	Export reviewed; live production status and public URL not supplied
Test status	Test cases specified in this report; execution evidence pending
Prepared by	SELVAGANAPATHY T - 241501194
Course	CB23F35 – Low-Code Application Development using Zoho Creator
Institution	Rajalakshmi Engineering College

TABLE OF CONTENTS

ABSTRACT
PROJECT AT A GLANCE
LIST OF TABLES
LIST OF FIGURES
LIST OF ABBREVIATIONS

CHAPTER 1 – INTRODUCTION
1.1 Background
1.2 Problem Statement
1.3 Objectives
1.4 Scope of the Project
1.5 Justification for a Low-Code Approach
1.6 Organisation of the Report

CHAPTER 2 – LITERATURE SURVEY AND SYSTEM STUDY
2.1 Study of the Existing System
2.2 Limitations of the Existing System
2.3 Low-Code versus Traditional Development
2.4 Literature Survey
2.5 Emerging Trends in Low-Code Development
2.6 Proposed System

CHAPTER 3 – REQUIREMENT ANALYSIS
3.1 Stakeholders and User Roles
3.2 Functional Requirements
3.3 Non-Functional Requirements
3.4 Hardware and Software Requirements
3.5 Use Case Model

CHAPTER 4 – SYSTEM DESIGN
4.1 System Architecture
4.2 Data Model
4.3 Form Design
4.4 Process and Workflow Design
4.5 User Interface and Navigation Design
4.6 Access Control Design

CHAPTER 5 – IMPLEMENTATION
5.1 Application Setup
5.2 Forms and Data Capture
5.3 Reports
5.4 Pages and Dashboards
5.5 Deluge Scripting
5.6 Workflows and Automation
5.7 Integrations
5.8 User Management and Role-Based Access
5.9 AI and Analytics Features
5.10 Pro-Code Extension
5.11 Mobile and Portal Access

CHAPTER 6 – TESTING AND DEBUGGING
6.1 Testing Strategy
6.2 Test Cases
6.3 Debugging Deluge Scripts
6.4 Defect Log
6.5 User Acceptance Testing

CHAPTER 7 – DEPLOYMENT, SECURITY AND GOVERNANCE
7.1 Environments and Deployment Strategy
7.2 Performance Optimisation
7.3 Security Features
7.4 Governance and Administration

CHAPTER 8 – RESULTS AND DISCUSSION
8.1 Application Output
8.2 Achievement of Objectives
8.3 Comparison with the Existing System
8.4 Limitations of the Low-Code Approach
8.5 Reflection on the No-Code → Low-Code → Pro-Code Spectrum

CHAPTER 9 – CONCLUSION AND FUTURE ENHANCEMENTS
9.1 Conclusion
9.2 Future Enhancements
9.3 Learning Outcomes and Career Relevance

REFERENCES

APPENDIX A – ZOHO CREATOR FEATURE COVERAGE CHECKLIST
APPENDIX B – COURSE OUTCOME MAPPING
APPENDIX C – DELUGE SOURCE CODE
APPENDIX D – USER MANUAL
D.1 Sign in and Open the Application
D.2 Register a Customer
D.3 Record a Solar Installation
D.4 Maintain Equipment Inventory
D.5 Maintain Technician Records
D.6 Create and Update a Maintenance Request
D.7 Schedule Maintenance and Complete Checklist
D.8 Use Reports and Dashboard
D.9 Operational Cautions

APPENDIX E – EVALUATION RUBRIC
APPENDIX F – VIVA-VOCE RECORD SHEET

LIST OF TABLES

Table 2.1: Comparison of Low-Code and Traditional Development for the Proposed Application

Table 2.2: Summary of Literature Survey

Table 3.1: Stakeholders and User Roles

Table 3.2: Functional Requirements

Table 3.3: Non-Functional Requirements

Table 3.4: Hardware and Software Requirements

Table 3.5: Use Case Description – Solar Installation Management

Table 4.1: Entities and Relationships

Table 4.2: Field Specification – Solar Installation Form

Table 4.3: Role–Permission Matrix

Table 5.1: Reports Implemented

Table 5.2: Deluge Scripts Implemented

Table 5.3: Workflows Implemented

Table 5.4: Integrations Implemented

Table 6.1: Test Cases

Table 6.2: Defect Log

Table 6.3: User Acceptance Feedback

Table 7.1: Deployment Checklist

Table 7.2: Performance Optimisation Results

Table 7.3: Security Controls Implemented

Table 8.1: Mapping of Objectives to Outcomes

Table 8.2: Existing System versus Proposed Application

Table A.1: Feature Coverage Checklist

Table B.1: Mapping of Course Outcomes to Report Chapters

Table E.1: Evaluation Rubric for the Individual Project

Table E.2: Performance Level Descriptors

Table F.1: Viva-Voce Questions and Responses

Table F.2: Marks Summary

LIST OF FIGURES

Figure 2.1: As-Is Process Flow of the Existing System

Figure 3.1: Use Case Diagram of the Solar Installation Management System

Figure 4.1: System Architecture of the Solar Installation Management System

Figure 4.2: Entity–Relationship Diagram

Figure 4.3: Workflow / Process Flow for Solar Installation Management

Figure 4.4: Navigation Structure and Wireframe of the Application

Figure 5.1: Application Home and Navigation Sections

Figure 5.2: Solar Installation Form in the Form Builder

Figure 5.3: Solar Installation Form in Live Mode

Figure 5.4: Installation Management Report

Figure 5.5: Application Dashboard

Figure 5.6: Workflow / Blueprint Configuration

Figure 5.7: Connection Setup and API Response Output

Figure 5.8: Roles and Permission Profiles Configuration

Figure 5.9: Analytics and Reporting Features

Figure 5.10: Application on Mobile / Customer Portal

Figure 6.1: Deluge Debug Output / Log

Figure 7.1: Environments / Publish to Production

Figure 7.2: Audit Trail / Administration Settings

Figure 8.1: Scenario Step 1 – Customer Registration and Installation Request

Figure 8.2: Scenario Step 2 – Technician Assignment and Installation Scheduling

Figure 8.3: Scenario Step 3 – Installation Completion and Status Update

LIST OF ABBREVIATIONS

Abbreviation	Expansion
API	Application Programming Interface
CRUD	Create, Read, Update, Delete
ER	Entity–Relationship
FR	Functional Requirement
KPI	Key Performance Indicator
LCNC	Low-Code / No-Code
NFR	Non-Functional Requirement
RBAC	Role-Based Access Control
UI	User Interface
UAT	User Acceptance Testing

CHAPTER 1

INTRODUCTION

1.1 Background

Solar installation work extends beyond the initial site installation. An organisation must maintain customer details, capture the technical configuration of each solar system, track equipment and spare parts, coordinate technicians, respond to service issues, and schedule preventive maintenance. These activities share data: a service request belongs to a customer and may refer to an installation; a maintenance schedule is associated with a customer, installation and technician; and a checklist records the inspection outcome.

The supplied Zoho Creator export implements a structured application for these related activities. It combines visual form configuration, reports, a dashboard page and Deluge-based workflow logic. This project is therefore a practical example of low-code application development applied to a domain-specific business process.

1.2 Problem Statement

Without a central operational record, staff may need to search across spreadsheets, email and paper notes to determine the installed system configuration, available stock, responsible technician or next maintenance date. Such fragmentation can make service history incomplete and stock decisions inconsistent. The project aims to centralise these records and apply selected validation and notification rules in one application.

1.3 Objectives

ID	Objective	Evidence in the supplied export
O1	Maintain structured customer and installation records	Customer and Solar Installation forms
O2	Track equipment quantities and minimum stock thresholds	Equipment Inventory form and low-stock validation workflow
O3	Record technicians and their availability/specialisation	Technician form and grouped technician reports

O4	Capture and follow up maintenance requests	Maintenance Request form, status/priority fields and reports
O5	Plan recurring maintenance and record inspection results	Maintenance Schedule and Maintenance Checklist forms
O6	Provide an at-a-glance operational view	Dashboard KPIs, charts and quick-add links
O7	Automate selected rules and notifications	Four configured workflow definitions in the export

1.4 Scope of the Project

Included scope: master-data forms, solar installation details and warranty fields, inventory tracking, technician records, maintenance request handling, maintenance scheduling, inspection checklists, list/grouped reports, a dashboard, low-stock validation, attempted assignment of available technicians for Critical requests, and email notifications on customer and technician creation.

Excluded or not evidenced: verified production deployment, live user roles and permission matrices, mobile/portal publication, integration with external CRM or APIs, scheduled reminder delivery, automated warranty-expiry alerts, payment collection, AI/Zia functionality, and measured performance results. These are not assumed to exist simply because Zoho Creator can support them.

1.5 Justification for a Low-Code Approach

Zoho Creator provides visual form and report builders and permits event-triggered Deluge scripting for rules that go beyond field configuration. This suits a project with several related business entities and a small number of targeted automations. The platform reduces the need to implement standard data-entry screens and reporting infrastructure from scratch, while still allowing logic such as stock validation and conditional technician assignment. Low-code does not remove the need for requirement analysis, data modelling, testing, access-control design or governance.

1.6 Organisation of the Report

Chapter 2 reviews the existing process and relevant literature. Chapter 3 defines stakeholders and requirements. Chapter 4 documents the architecture, data model and process design. Chapter 5 describes the implementation evidenced in the export.

Chapters 6–8 cover testing, deployment/security and results. Chapter 9 concludes with limitations and future enhancements. The appendices provide a feature checklist, course-outcome mapping, code excerpts and operating guidance.

CHAPTER 2

LITERATURE SURVEY AND SYSTEM STUDY

2.1 Study of the Existing System

A typical manual or spreadsheet-based solar service process starts with a customer enquiry or installation record. Installation details are recorded separately; inventory is maintained in a stock sheet; technicians are contacted through calls or messages; service visits are scheduled in calendars; and inspection results are kept in paper checklists or separate documents. The following figure represents a conceptual as-is workflow and should be replaced with the actual process followed by the organisation, if different.

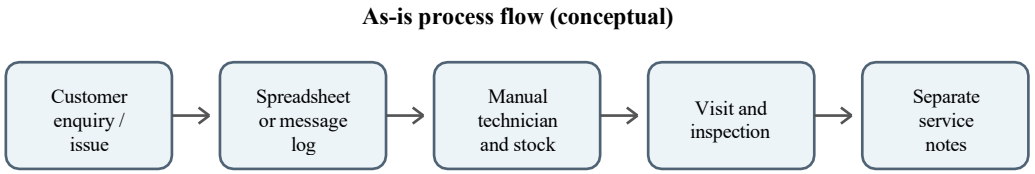


Fig. 2.1 As-is process flow (conceptual; validate against actual practice).

2.2 Limitations of the Existing System

Limitation	Operational effect	Response in this project
Fragmented customer/install records	Staff may need to search multiple files	Customer and Solar Installation forms
Inventory updated separately	Low stock may be overlooked	Minimum-stock field and validation workflow
Technician availability checked manually	Assignment may be delayed	Availability field and Critical-request assignment workflow
Service details not consistently structured	Diagnosis/action history is difficult to compare	Maintenance Request and Checklist forms

No consolidated overview	Managers lack a quick summary	Dashboard KPI tiles and charts
Manual follow-up reminders	Preventive service may be missed	Schedule records exist; automated reminders are not evidenced

2.3 Low-Code versus Traditional Development

Criterion	Traditional pro-code	Low-code on Zoho Creator
Form/report construction	Custom UI and database code typically required	Visual form and report configuration
Business logic	General-purpose language and framework	Deluge scripts plus built-in actions
Iteration	Requires coordinated code changes and deployment	Many layout/configuration changes can be made in the builder
Portability	Depends on chosen language, framework and infrastructure	Tied to Zoho Creator data model/runtime unless migrated
Governance	Designed through application code and infrastructure controls	Must be configured through Creator permissions, roles and platform administration
Complexity ceiling	Broad control over architecture	Platform boundaries and plan-dependent features apply

2.4 Literature Survey

Source	Key finding	Relevance
Bucaioni, Cicchetti and Ciccozzi (2022) [1]	Systematic review of low-code modelling and its relation to model-driven engineering.	Supports describing visual models and configuration as a core part of low-code.
Pinho, Aguiar and Amaral (2023) [2]	Reviews low-code/no-code research through a usability lens; highlights usability as a material concern.	Motivates clear forms, navigation and testing with intended users.
Gao, Fagerholm and Toivanen (2026) [4]	Review reports evidence that low-code can accelerate development	Supports a balanced evaluation and avoidance

	while quality/complexity outcomes depend on context.	of unmeasured benefit claims.
Zoho Creator Help: Form Workflows [5]	Describes record/form-event workflows and available actions including Deluge and notifications.	Direct platform reference for the configured workflows in this application.
Zoho Creator Help: User Permissions [6]	Explains permission, role and data-sharing controls for limiting access.	Used as a design reference; actual role configuration must still be verified.

2.5 Emerging Trends in Low-Code Development

Current low-code practice increasingly combines visual modelling with custom scripting, integrations, reusable components, automation and analytics. These capabilities can shorten the path from a business requirement to a working prototype, but they also create a need for controlled change management, maintainable logic and deliberate access control. The current application uses a restrained hybrid approach: most screens and reports are configured visually, while Deluge is used for selected validation, assignment and email actions.

2.6 Proposed System

The proposed system centralises records for customers, installations, equipment, technicians and service activity. The dashboard surfaces aggregate counts and capacity information. Reports support operational review by categories and statuses. The design improves data organisation and provides automation hooks, but real operational improvement must be demonstrated by testing and comparing the live process with a documented baseline.

CHAPTER 3

REQUIREMENT ANALYSIS

3.1 Stakeholders and User Roles

Stakeholder	Description	Responsibilities / needs
Administrator / app owner	Configures the Creator application	Maintain forms, workflows, reports, users and platform settings
Operations manager	Oversees installation and service workload	Review installation status, requests, schedules, stock and dashboard
Service coordinator	Receives and organises service requests	Register requests, schedule visits, track status and assigned technician
Technician	Performs installation/maintenance work	Review assigned work, record diagnosis, action taken and checklist results
Inventory coordinator	Maintains equipment availability	Record stock movement and respect minimum-stock threshold
Customer	Recipient of installation and maintenance service	Provide correct contact/site details; customer portal is not evidenced

These are conceptual user roles for requirements analysis. The .ds export defines application modules but does not prove that matching role profiles or row-level permissions have been configured. Verify the actual live user-permission settings before asserting role-based access.

3.2 Functional Requirements

ID	Requirement	Priority	Evidence / status
FR-01	Create and maintain customer name, phone, email, address and customer type.	High	Customer form
FR-02	Register installations with capacity, panel count/type, inverter details, installation date and warranty fields.	High	Solar Installation form
FR-03	Maintain equipment category, quantity, minimum stock, unit and storage location.	High	Equipment Inventory form

FR-04	Prevent inventory submission when quantity available is below minimum stock level.	High	Configured on-validate workflow; live execution to test
FR-05	Maintain technician contact details, specialisation and availability.	High	Technician form
FR-06	Register maintenance requests with issue category, priority, status, description and service notes.	High	Maintenance Request form
FR-07	For Critical requests, attempt to assign an available technician; fall back to the logged-in user if they are listed as a technician.	High	Configured on-success workflow; test lookup/assignment behavior
FR-08	Maintain preventive maintenance schedules with interval, due date and status.	Medium	Maintenance Schedule form
FR-09	Capture checklist condition, observations and corrective action.	Medium	Maintenance Checklist form
FR-10	View reports grouped by key operational categories.	Medium	Grouped/kanban reports
FR-11	View dashboard KPIs, charts and quick-add actions.	Medium	DashBoard page
FR-12	Send a customer registration email after customer creation.	Medium	Configured email workflow; live delivery to test
FR-13	Send a technician-related email after technician creation.	Medium	Configured email workflow; verify template content and delivery

3.3 Non-Functional Requirements

ID	Category	Requirement / acceptance criterion
NFR-01	Usability	Field labels should be understandable to operations staff; confirm through UAT.
NFR-02	Data integrity	Related records must reference the intended customer, installation, technician and request.
NFR-03	Validation	The stock threshold rule should block invalid submissions and explain the reason.

NFR-04	Reliability	Critical-request assignment and email actions should handle missing matches or unavailable recipients gracefully.
NFR-05	Security	Only authorised users should access and modify operational data; configure and test actual permissions.
NFR-06	Performance	Record counts and report load times should be measured in the live app; no measured target/result is supplied.
NFR-07	Maintainability	Use clear field/link names and docu2ment workflow purpose and trigger.
NFR-08	Auditability	Record creation, modification and workflow outcomes should be reviewable using available platform audit/history features.

3.4 Hardware and Software Requirements

Category	Specification
Platform	Zoho Creator cloud application; exact edition/plan not identified by the export
Scripting	Deluge, as used in exported form workflows
Client device	Desktop/laptop with a supported modern browser; mobile browser support should be verified
Connectivity	Internet connection required for cloud application access
Authoring input	Solar_Installation_Management_System.ds export
Evidence needed	Live app URL, platform edition, browser/version, test account roles and screenshots

3.5 Use Case Model

Use-case overview

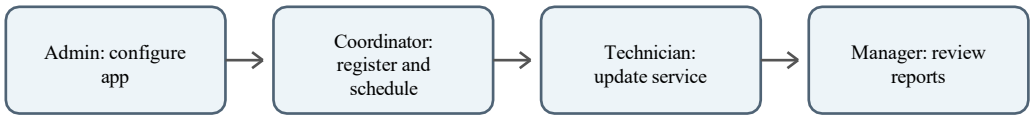


Fig. 3.1 Use-case overview (conceptual actors; permissions need live verification).

Use Case ID	UC-01
Use Case Name	Register a maintenance request
Actor(s)	Service coordinator / authorised staff
Precondition	Customer and, where applicable, installation records exist.
Main flow	Open Maintenance Request; select customer/installation; enter issue, priority and dates; submit; inspect request report.
Alternate flow	If priority is Critical, configured workflow attempts technician assignment.
Postcondition	Request record is stored; assignment and notification behavior must be confirmed by live test.

CHAPTER 4

SYSTEM DESIGN

4.1 System Architecture

The export indicates a cloud-hosted Zoho Creator application composed of form-based data capture, reports, a custom dashboard page and form workflows. Deluge executes selected business rules in response to record events. The exported file is a configuration representation, not proof of a deployed endpoint or running data set.

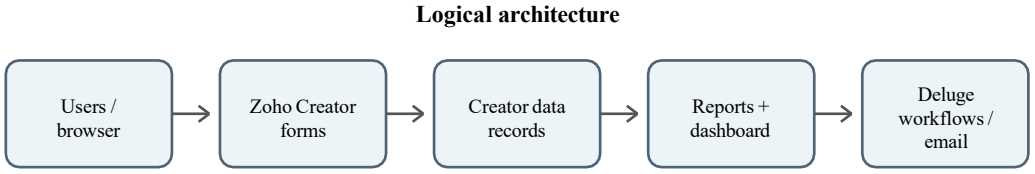


Fig. 4.1 Logical system architecture derived from the exported configuration.

4.2 Data Model

Entity (form)	Key fields evidenced	Relationships / notes
Customer	customer_name, phone_number, email, address, customer_type, customer_code	Referenced by installation, maintenance request and schedule records through lookup/list-type fields.
Solar Installation	installation_name, customer, system_capacity_kw, panel_count, panel_type, inverter_model, warranty_expiry_date, installation_status	Belongs to a customer; referenced by maintenance requests and schedules.
Equipment Inventory	equipment_name, equipment_category, quantity_available, minimum_stock_level, stock_status	Independent stock records; no consumption transaction form is evidenced.
Technician	technician_name, email, specialization, availability_status, technician_code	Referenced by maintenance requests and schedules as assigned technician.

Maintenance Request	customer, installation, issue_category, priority, status, assigned_technician, diagnosis, action_taken	Connects customer/install context to technician and service notes.
Maintenance Schedule	customer, installation, maintenance_type, maintenance_interval_months, next_due_date, assigned_technician	Represents planned maintenance; automatic reminders are not evidenced.
Maintenance Checklist	checklist_item, condition_1, observation, corrective_action, maintenance_request	Checklist entry is associated with a maintenance request.

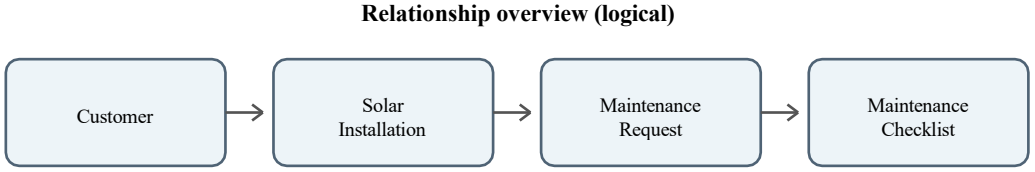


Fig. 4.2 Logical relationship overview. The exact lookup cardinality should be checked in the live Creator builder.

4.3 Form Design

Form	Purpose	Notable field types / validation
Customer	Customer master	Text, phone, email, address with coordinates, picklist, date, textarea
Solar Installation	System/site configuration	Lookup/list, address, decimal capacity, number of panels, panel-type picklist, dates, warranty fields
Equipment Inventory	Stock and threshold tracking	Text, picklists, numeric quantity/minimum, textarea; low-stock validation workflow
Technician	Technician directory	Text, phone, email, specialisation and availability picklists, date, textarea
Maintenance Request	Service intake and resolution	Customer/installation references, dates, issue and priority/status picklists, textareas, assigned technician

Maintenance Schedule	Preventive maintenance plan	Customer, installation, maintenance type, interval number, dates, status, notes, assigned technician
Maintenance Checklist	Inspection outcome	Checklist item and condition picklists, observations/corrective action textareas, maintenance-request reference

4.4 Process and Workflow Design

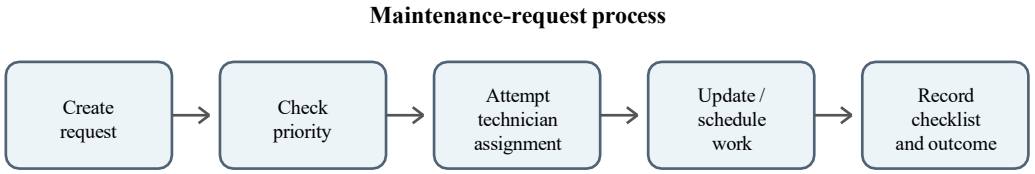


Fig. 4.3 Maintenance-request process derived from available forms and workflow logic; steps after assignment are operational guidance, not a fully automated blueprint.

Workflow WF-01 (Auto-Assign Technician for High Priority) runs after a Maintenance Request is added. The script checks whether the request priority equals "Critical", fetches technicians whose availability status is "Available", and assigns the first returned technician if the result count is greater than zero. If none is available, it searches for a technician whose email matches the logged-in user. The actual field assignment semantics and result-set selection should be tested in Creator.

Workflow WF-02 (Prevent Low Stock Submission) runs on validation for Equipment Inventory on add or edit. If quantity_available is below minimum_stock_level, it alerts the user and cancels submission. WF-03 sends a customer registration email after a customer is added. WF-04 sends an email after a technician is added; the configured message contains an employment-offer-style template and references "ABC solars", so it should be reviewed and rewritten to match the actual business purpose before use.

4.5 User Interface and Navigation Design

The Dashboard page contains KPI cards for average system capacity, total maintenance requests, total solar installations and total maintenance schedules. It also contains quick-add panels, a scatter chart relating system capacity to panel count, a bar chart for panel type by panel count, and a maintenance request report component.

Navigation entries exist for the seven primary forms and their standard/grouped reports.

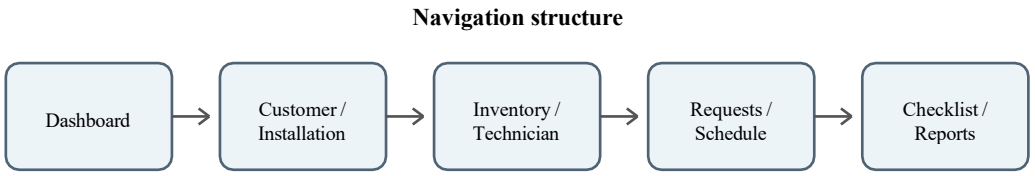


Fig. 4.4 Navigation structure derived from the exported page and menu configuration.

4.6 Access Control Design

The report proposes least-privilege access: administrators maintain configuration; coordinators manage customer and service records; technicians view and update assigned service activity; inventory staff update stock; managers view aggregate reports. The export does not demonstrate that these roles or row-level permissions are actually configured. Record and report access, export rights, deletion rights, sharing settings and any portal access must be inspected and tested in the live application before this design can be claimed as implemented.

CHAPTER 5

IMPLEMENTATION

5.1 Application Setup

The supplied file identifies the application as “Solar Installation Management System”, version 1.0, generated on 21-Oct-2026 19:34:07. It declares a date format of dd-MMM-yyyy, a 24-hour time format and the timezone America/Los_Angeles. Because the application is intended for an Indian academic context, confirm that the configured timezone matches the intended operating region before live use.

The application is organised around seven forms, reports and a page named DashBoard.

The export was used as the implementation evidence source for this report. The exact Creator plan, live app link, owner account, environment separation and production deployment date were not provided.

5.2 Forms and Data Capture

The seven forms cover customer master data, installation specifications, equipment stock, technician profiles, maintenance intake, preventive schedules and inspection checklists. Across the forms, the export shows multiple distinct field types including text, phone, email, address, picklist, date, textarea, number, decimal, radio buttons and list/lookup-style references. Customer and installation addresses capture coordinates, although no map-based routing feature is demonstrated.

5.3 Reports

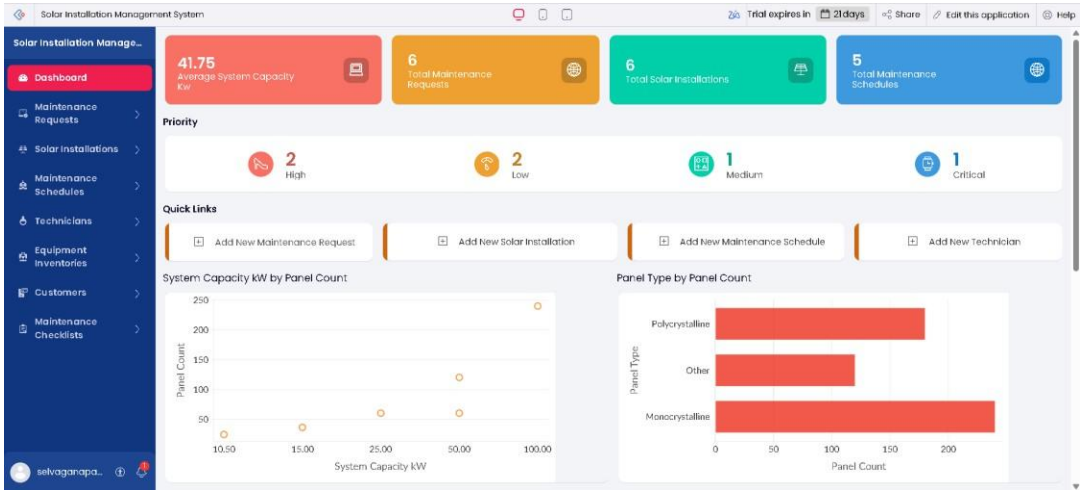
Source form	Reports evidenced in export
Maintenance Request	maintenance_request_Report; maintenance_request_by_issue_category
Solar Installation	solar_installation_Report; solar_installation_by_installation_status
Maintenance Schedule	maintenance_schedule_Report; maintenance_schedule_by_maintenance_type
Technician	technician_Report; technician_by_specialization

Equipment Inventory	equipment_inventory_Report; equipment_inventory_by_equipment_category
Customer	customer_Report; customer_by_customer_type
Maintenance Checklist	maintenance_checklist_Report; maintenance_checklist_by_checklist_item

The export defines both standard report views and grouped/kanban views. Filters, sorting, record-level actions and visible columns should be verified in live mode. The export alone does not provide data volume or actual report response time.

5.4 Pages and Dashboards

The DashBoard page contains four summary metrics: average system capacity in kW, total maintenance requests, total solar installations and total maintenance schedules. Two charts are configured: a scatter chart with system capacity and panel count, and a bar chart summarising panel count by panel type. The page also includes quick-add links for maintenance requests, installations, maintenance schedules and technicians, and embeds a maintenance-request report.



. *Fig. 5.1 Dashboard screenshot — pending insertion from live Zoho Creator.*

5.5 Deluge Scripting

Script ID	Location / trigger	Purpose	Concepts
DS-01	Maintenance Request → On Success, On Add	For Critical priority, attempt assignment to an available technician; fallback to logged-in user if they are a technician.	Record fetch, conditional branch, count, field assignment
DS-02	Equipment Inventory → On Validate, On Add or Edit	Cancel submit when available quantity is below minimum stock.	If, alert, cancel submit
DS-03	Customer → On Success, On Add	Send registration email and attach configured PDF template.	sendmail, field interpolation, template attachment
DS-04	Technician → On Success, On Add	Send email based on a configured template.	sendmail, field interpolation, template attachment

Important implementation review: the high-priority workflow assigns `firstTechnician = availableTechnicians` rather than visibly selecting one record by a deterministic sort. Verify that Creator accepts the returned collection in the assignment context; if not, explicitly select one record and define a deterministic ordering. Also test the case where the logged-in technician lookup returns multiple records or none. The technician email body appears to contain a hiring/offer template unrelated to a technician record being added; revise it before production.

5.6 Workflows and Automation

Workflow	Trigger	Action	Verification needed
Auto-Assign Technician for High Priority	Maintenance Request: On Add / On Success	If priority is Critical, search Available technicians and assign;	Test no available tech, one available tech, multiple techs,

		fallback search by logged-in email	non-Critical request and fallback
Prevent Low Stock Submission	Equipment Inventory: On Add or Edit / On Validate	Alert and cancel when available quantity < minimum stock	Test below/equal/above threshold and numeric blanks
Notify Customer	Customer: On Add / On Success	Send welcome/registration email with PDF template attachment	Verify sender permissions, address, template and delivery
Technician appointed	Technician: On Add / On Success	Send email and PDF template attachment	Review business wording and verify delivery

5.7 Integrations

No external REST API, CRM sync, payment gateway or third-party integration call is evidenced in the supplied export. The two email workflows use Zoho Creator's sendmail action, which is a platform notification capability. Email sending depends on account configuration, sender permissions, template correctness and recipient validity; it must be verified by an actual test.

5.8 User Management and Role-Based Access

The supplied configuration contains application navigation and components but does not provide enough evidence to assert a completed role/permission matrix. The student should capture the Creator user-permissions page and run negative access tests using separate accounts for administrator, coordinator and technician roles if those roles are configured.

5.9 AI and Analytics Features

No AI/Zia feature is evidenced. Analytics is limited to the configured dashboard KPI cards and two charts. The scatter chart uses system_capacity_kw and panel_count; the bar chart aggregates panel_count by panel_type. Values depend on live records, so screenshots should use clearly identified test data and should not imply real customer/installation totals.

5.10 Pro-Code Extension

A separate pro-code extension was not used. The application uses Creator's visual builders and Deluge workflow scripting; no custom external server, independent database service or custom front-end code is evidenced in the export. This is appropriate to the scope of a low-code coursework project, provided the limitations and platform dependency are documented.

5.11 Mobile and Portal Access

The export does not demonstrate a published mobile application, customer portal or external-facing portal. Zoho Creator may offer responsive access depending on configuration, but actual mobile layout, portal permissions and user experience must be tested on the target devices before being claimed. Insert a genuine mobile screenshot only if the application has been tested on a mobile device.

CHAPTER 6

TESTING AND DEBUGGING

6.1 Testing Strategy

Testing should be performed in a development or test copy before production. The cases below are a test plan derived from the requirements and workflow configuration; they are not reported as executed. Fill the Actual Result and Status columns only after running each case in the live Creator application and retaining screenshotsorexecutionnotes.

6.2 Test Cases

TC ID	FR Ref.	Test description / input	Steps	Expected result	Actual / Status
TC-01	FR-01	Create customer with valid name, phone and email	Open Customer; enter valid fields; submit	Record saved and visible in Customer report	Pending execution
TC-02	FR-01	Customer email field receives malformed email	Enter invalid email; submit	Platform validation rejects or flags invalid email	Pending execution
TC-03	FR-02	Create installation with valid capacity and panel count	Select customer; complete required fields; submit	Installation saved and linked to customer	Pending execution
TC-04	FR-02	Enter zero or negative system capacity	Enter boundary/invalid capacity; submit	Business rule should reject invalid value if such validation is configured; verify current behavior	Pending execution
TC-05	FR-03	Create equipment with quantity	Set quantity 10, minimum 5; submit	Record saves	Pending execution

		above minimum			
TC-06	FR-04	Quantity below minimum	Set quantity 4, minimum 5; submit	Alert appears and submit is cancelled	Pending execution
TC-07	FR-04	Quantity equal to minimum	Set quantity 5, minimum 5; submit	Current condition (<) does not block equality; record should save	Pending execution
TC-08	FR-04	Edit existing equipment to below minimum	Edit quantity from 8 to 3, minimum 5; save	Alert appears and edit is cancelled	Pending execution
TC-09	FR-05	Create technician marked Available	Complete technician details; submit	Technician record appears in report	Pending execution
TC-10	FR-07	Create Critical request with an Available technician	Create request with priority Critical	Workflow assigns an available technician if lookup and assignment work	Pending execution
TC-11	FR-07	Create Critical request with no Available technician and logged-in user is a technician	Mark all technicians unavailable; login as technician; create request	Workflow attempts fallback assignment to matching logged-in email	Pending execution
TC-12	FR-07	Create High/Medium request	Create request with priority other than Critical	Auto-assignment branch should not execute	Pending execution
TC-13	FR-06	Create request without optional installation	Create request for customer only if form allows it	Record behavior matches configured field requirement and relationships	Pending execution
TC-14	FR-08	Create schedule with	Enter maintenance type,	Schedule saved and appears in report	Pending execution

		interval and next due date	interval, dates and customer		
TC-15	FR-09	Record checklist with Needs Attention condition	Choose checklist item and condition; add observation/action	Checklist record saves and grouped report reflects condition	Pending execution
TC-16	FR-12	Create customer and inspect email	Submit customer using controlled test mailbox	Registration email arrives with correct interpolation and attachment	Pending execution
TC-17	FR-13	Create technician and inspect email	Submit technician using controlled test mailbox	Email content is relevant and delivered; template wording reviewed	Pending execution
TC-18	FR-10	Open grouped reports	Open each grouped report and inspect grouping	Records group under issue/status/ type/specialisation/category/type as intended	Pending execution
TC-19	FR-11	Dashboard KPI and charts with known test records	Insert controlled records; open dashboard	KPI values/charts reflect the test data	Pending execution
TC-20	NFR-05	Attempt access using a restricted test user	Login with least-privilege account; open restricted report/form	Unauthorised access is denied according to configured permissions	Pending execution
TC-21	NFR-06	Measure report loading time with documented record count	Repeat timed load under stable conditions	Record count and measured response time documented; compare with chosen target	Pending execution
TC-22	NFR-07	Edit and cancel a record update	Open existing record; change fields; cancel	No unintended changes are persisted	Pending execution
TC-23	FR-02	Warranty expiry date review	Enter installation and warranty values	Stored values display correctly; automatic expiry alert is not assumed	Pending execution

TC-24	FR-06	Request completion fields	Update status and completion date; enter diagnosis/action	Record stores service outcome and appears in report	Pending execution

6.3 Debugging Deluge Scripts

For each workflow, reproduce the triggering record event with controlled data, inspect the Creator execution/error details available for that workflow, confirm the criteria matched the intended records, and check whether field assignment and email actions completed. Record the exact failing input, timestamp, error text, root cause and correction. The low-stock rule should be tested with values below, equal to and above the threshold. The assignment rule should be tested with zero, one and multiple available technicians.

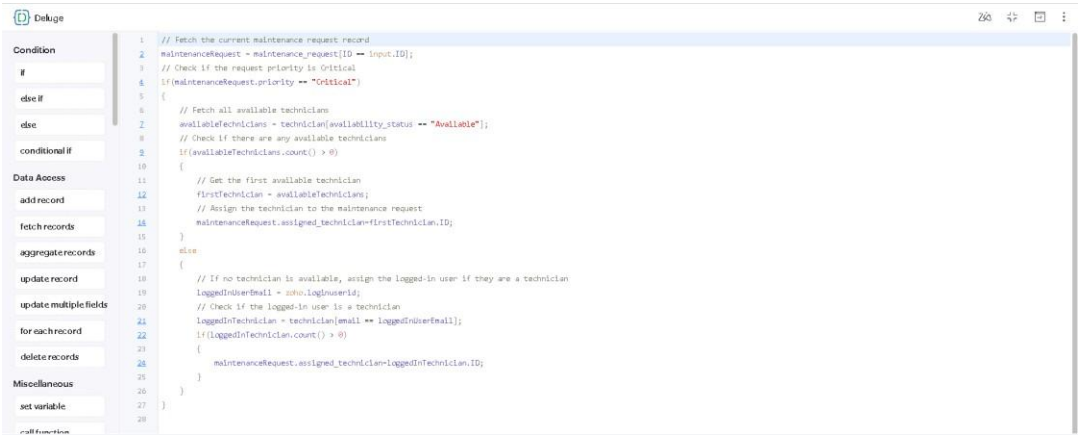


Fig. 6.1 Deluge debug evidence — pending live execution.

6.4 Defect Log

Bug ID	Description	Root cause	Fix applied	Status
BUG-01	Potential nondeterministic technician selection / collection assignment	Script assigns the availableTechnicians result to firstTechnician without explicit ordering/selection	Verify in live Creator; if required select one record explicitly and define order	Open / verify

BUG-02	Technician email text appears to be an employment offer unrelated to record creation	Copied or mismatched message template in workflow	Replace with an accurate technician onboarding/record confirmation message	Open / recommended
BUG-03	Application timezone is America/Los_Angeles	Export setting may not match intended operating region	Confirm business timezone and change if needed after stakeholder approval	Open / verify

6.5 User Acceptance Testing

UAT has not been evidenced by the export. After testing, record each tester's name or role (with permission), the scenario exercised, feedback, action taken and sign-off date. Do not fabricate participant names or claim acceptance before an actual review.

CHAPTER 7

DEPLOYMENT, SECURITY AND GOVERNANCE

7.1 Environments and Deployment Strategy

Recommended deployment sequence: (1) retain the .ds export as a versioned baseline; (2) import/configure the app in a development environment; (3) verify fields, relationships, workflows, email templates and reports; (4) execute the test cases with controlled data; (5) correct defects and rerun regression tests; (6) verify user permissions; (7) publish or share the application with intended users; and (8) retain a deployment checklist and rollback/export copy. The supplied file does not identify a live production URL or prove that these steps have been completed.

Checklist item	Status based on supplied evidence
Application export received and reviewed	Yes
Live app URL recorded	Not supplied
Development/stage environment confirmed	Not evidenced
Test cases executed	Pending
Production publish/deploy completed	Not evidenced
User access and permission tests completed	Pending
Backup/rollback copy retained	Export file supplied; backup policy not evidenced

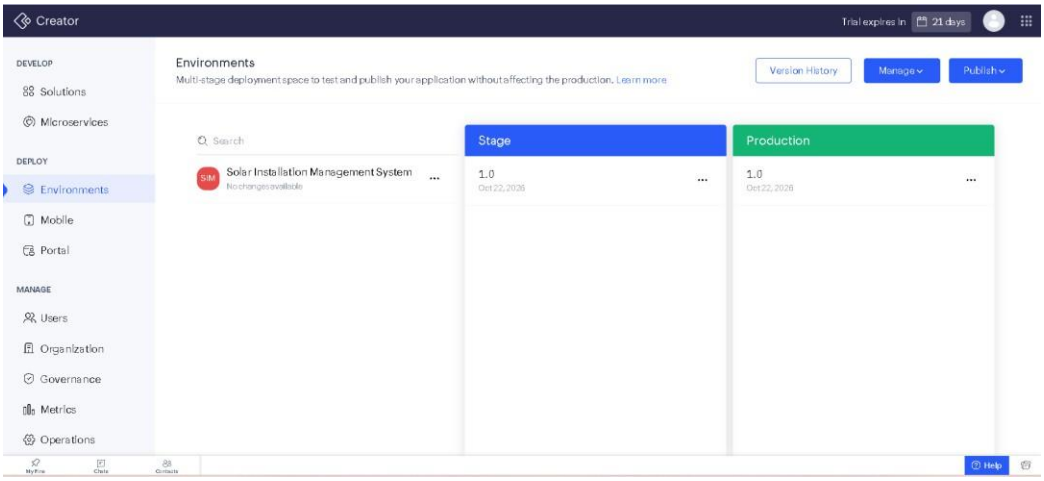


Fig. 7.1 Deployment evidence — pending live capture.

7.2 Performance Optimisation

No benchmark data is supplied. Potential improvements include restricting reports to useful default filters, avoiding unnecessary record fetches inside loops, selecting a deterministic technician candidate, limiting dashboard components to operationally relevant summaries and monitoring workflow execution. Any before/after performance claim should include record count, measurement method and observed timings.

7.3 Security Features

Security control	Application relevance	Current evidence / action
Authentication	Ensure only approved users can open the application	Verify account/user-sharing settings in live Creator
Least privilege	Limit editing/deletion/export according to job role	Permission matrix not evidenced; configure and test
Data minimisation	Collect only customer/site data needed for service	Review phone, email, address and location fields
Email correctness	Prevent misdirected messages or incorrect content	Validate recipient fields and revise technician template
Input validation	Prevent invalid stock state	Low-stock validation is configured; test negative/equal/above cases
Audit/recovery	Support traceability and recovery from mistakes	Confirm available audit history and export/backup procedure
Data sharing	Avoid broad report sharing or public exposure	Public portal/sharing settings not evidenced; inspect before release

7.4 Governance and Administration

A production-ready governance process should identify the app owner, approvers for changes, naming conventions, a release log, periodic access reviews, backup/export frequency, incident response and retention rules. Deluge changes should be documented with trigger, purpose, expected result and regression tests. Since the current export uses a US timezone and includes a technician email template with

unrelated hiring language, both configuration and content review should be part of release readiness.

CHAPTER 8

RESULTS AND DISCUSSION

8.1 Application Output

The supplied export confirms that the application configuration includes seven primary forms, a dashboard page, standard reports and grouped/kanban reports, and four workflow definitions. It does not contain screenshots of the running application, live sample records, email delivery logs, performance timings or a deployment URL. Results are therefore reported as configuration-level evidence rather than measured operational outcomes.

Output area	Configuration-level result	Evidence still needed
Customer and installation records	Forms and related fields are defined	Live-mode form screenshots and saved test records
Inventory validation	On-validate low-stock rule is configured	Below/equal/above threshold test evidence
Technician assignment	Critical-priority assignment logic is configured	Test evidence for available/no-available/fallback scenarios
Email notification	Two sendmail actions are configured	Successful delivery, final content and attachment evidence
Reports	Standard and grouped reports are defined for the seven forms	Live screenshots with controlled sample data
Dashboard	KPI tiles, charts and quick-add links are configured	Screenshot and cross-check of displayed values
Security/deployment	Not established by the export alone	Live permissions, URL and publish configuration

8.2 Achievement of Objectives

Objective	Implementation evidence	Assessment
O1: Maintain customer/install records	Customer and Solar Installation forms	Configuration evidenced; live data capture to verify
O2: Track stock and threshold	Equipment Inventory form and validation workflow	Configured; negative and boundary tests pending

O3: Maintain technician information	Technician form and specialization report	Configuration evidenced
O4: Capture maintenance requests	Maintenance Request form and reports	Configuration evidenced; end-to-end handling pending
O5: Plan maintenance and capture checks	Maintenance Schedule and Checklist forms	Configuration evidenced; reminders not evidenced
O6: Provide operational overview	Dashboard KPIs, charts and quick links	Page configuration evidenced; values need live validation
O7: Automate rules/notifications	Four workflow definitions	Configured; live execution and delivery pending

8.3 Comparison with the Existing System

Compared with disconnected spreadsheets and messages, the proposed design places related operational data in a single Creator application and standardises key fields. The dashboard can aggregate counts and capacity values, while grouped reports make records easier to inspect by category or status. However, no baseline timings, user-study results or actual record volumes were supplied; therefore, percentage improvement, time saved and error reduction cannot be quantified honestly at this stage.

8.4 Limitations of the Low-Code Approach

The application depends on Zoho Creator's platform, data model and workflow runtime. Portability to another platform would require mapping fields, rebuilding reports and translating Deluge logic. The quality of the outcome also depends on correct relationship definitions, data quality, permission configuration and testing. The current export reveals several areas to review: nondeterministic technician selection, the technician email template content and the configured timezone.

8.5 Reflection on the No-Code → Low-Code → Pro-Code Spectrum

Most of the application is configured through visual form/report/page builders, which falls toward the no-code end of the spectrum. The stock validation, conditional assignment and email actions use Deluge and therefore represent low-code scripting.

A separate pro-code extension is not present in the supplied export. This combination is proportionate to the project, while more complex integrations, advanced scheduling or high-volume transaction logic may require additional platform capabilities or external services.

CHAPTER 9

CONCLUSION AND FUTURE ENHANCEMENTS

9.1 Conclusion

The Solar Installation Management System is a Zoho Creator low-code application configuration for coordinating customer, installation, inventory, technician and maintenance information. The supplied export shows seven primary forms, a dashboard with KPIs and charts, grouped reports, and four configured workflows. The most concrete rule is the low-stock validation that is intended to stop a submission when available quantity is lower than the minimum stock level. The Critical-request workflow attempts to assign an available technician, and customer/technician creation workflows are configured to send emails.

The report deliberately distinguishes these configured features from proven runtime outcomes. The export alone cannot establish that the application is deployed, that emails were delivered, that assignment logic behaves correctly under all cases, that permissions are secure, or that users achieved measurable time savings. Completing the live test plan, reviewing the workflow details and capturing genuine screenshots are necessary steps before final academic submission.

9.2 Future Enhancements

- Correct and validate technician assignment by selecting one deterministic candidate andhandling zero/multiple matches.
- Replace the technician email template with a relevant record-creation or onboardingnotification.
- Confirm the timezone and date-handling rules for the intended deployment region.
- Add automated reminders for upcoming maintenance and warranty expiry, withduplicate-notification protection.
- Introduce stock issue/consumption transactions linked to maintenance requests rather thanmaintaining only a current quantity.
- Add approval stages or a Blueprint for request lifecycle transitions if the business requirescontrolled status changes.

- Configure and test role-based permissions and record-level access for coordinators,technicians and managers.
- Add dashboards for overdue schedules, critical open requests, low-stock items and warrantyexpiries.
- Evaluate customer/technician portal access and mobile usability, with carefully limited dataexposure.
- Create a test data set, regression suite, release checklist and operational backup/export policy.

9.3 Learning Outcomes and Career Relevance

The project demonstrates domain analysis, relational data modelling, visual form design, report configuration, dashboard composition and event-driven scripting. It also highlights professional responsibilities beyond feature creation: validating edge cases, ensuring message content is appropriate, checking timezone assumptions, securing data access and presenting evidence-based results. These practices are relevant to business application development, workflow automation, systems analysis and low-code solution delivery.

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[5] Zoho Corporation, “Creating and managing form workflows,” Zoho Creator Help, https://help.zoho.com/portal/en/kb/creator/developer-guide/workflows/create-and-manage-form-workflo ws (accessed Oct. 21, 2026).

[6] Zoho Corporation, “Understanding user permissions,” Zoho Creator Help, https://help.zoho.com/portal/en/kb/creator/developer-guide/application-settings/manage-and-configure-user-per missions/articles/understand-user-permissions (accessed Oct. 21, 2026).

[7] Zoho Corporation, Solar Installation Management System application export supplied forthis report, file: Solar_Installation_Management_System.ds, version 1.0, generated Oct. 21, 2026.

APPENDIX A – ZOHO CREATOR FEATURE COVERAGE CHECKLIST

Checklist statuses below are based only on the supplied .ds export. “Configured” means the feature is represented in the export, not that it has been executed successfully in a live app.

Items requiring live evidence are marked “Verify”.

Unit	Feature / checklist item	Status	Where evidenced	Report section
I	Application created and sections/menu organised	Configured	Application name, forms, reports and navigation	5.1
I	Forms with at least 8 distinct field types	Configured	Text, phone, email, address, picklist, date, textarea, number, decimal, radio/list fields	5.2
I	Form field properties / mandatory rules checked	Verify	Field definitions exist; full live behavior not tested	5.2 / 6
I	Reports with filtering/grouping/sorting	Configured in export	Standard and grouped/kanban report definitions	5.3
I	Pages / dashboard designed	Configured	DashBoard page	5.4
I	Lookup/relationship fields used	Configured in export	Customer, installation, technician and request references	4.2
II	Workflow triggered by form event	Configured	Four form workflows	5.6
II	Deluge script with conditional logic	Configured	Critical assignment and low-stock validation	5.5
II	Data validation and error handling	Configured	Stock validation alert and cancel submit	5.5
II	Email notification	Configured	Customer and technician sendmail actions	5.6

II	Workflow execution verified	Verify	No live execution log supplied	6.2
II	Scheduled workflow / recurring job	Not evidenced	No schedule workflow found in export	5.6
II	Approval process / Blueprint	Not evidenced	No approval/Blueprint definition identified	5.6
III	Integration with Zoho CRM or external API	Not evidenced	No external API/CRM action identified	5.7
III	Connections / OAuth configuration	Not evidenced	No connection setup shown	5.7
III	PDF template attachment in email	Configured reference	_default1 PDF template referenced in two email actions	5.6
III	AI/Zia feature	Not evidenced	No AI feature identified	5.9
III	Analytics/chart component	Configured	Scatter and bar chart on dashboard	5.4 / 5.9
IV	Role hierarchy / permission sets	Verify	Export does not prove actual permission settings	5.8 / 7.3
IV	Record-level access tested	Verify	No test evidence supplied	6.2
IV	Audit trail / administration settings reviewed	Verify	Live settings not supplied	7.4
IV	Production deployment/publish	Verify	Live URL/status not supplied	7.1
IV	Mobile/portal access tested	Not evidenced	No mobile/portal configuration demonstrated	5.11
V	Pro-code extension	Not used / not evidenced	No external server or custom front end	5.10
V	Performance benchmark	Pending	No record volume or timings supplied	7.2
V	User acceptance testing	Pending	No UAT record supplied	6.5

V	Test data isolated/removed	Pending	Live data environment not supplied	7.1
V	Backup/export retained	Partial	Source .ds export supplied; ongoing backup policy not evidenced	7.1
V	Documentation and user manual	Included	Appendix D	Appendix D
V	At least 15 test cases documented	Yes (planned)	24 test cases listed; execution pending	6.2
V	At least 5 literature sources	Yes	Seven references listed	References

APPENDIX B – COURSE OUTCOME MAPPING

CO	Course outcome statement	Evidence in project	Self-assessed level	Examiner level
CO1	Explain principles, architecture and business value of low-code application development using Zoho Creator.	Chapters 1, 2 and 4	[Student to assess]	—
CO2	Develop forms, reports, pages and data relationships in Zoho Creator.	Chapters 4 and 5; seven forms and reports	[Student to assess]	—
CO3	Implement business rules and automation using Deluge and workflows.	Chapters 5 and Appendix C	[Student to assess]	—
CO4	Test, debug and validate a low-code application.	Chapter 6; test plan pending execution	[Student to assess]	—
CO5	Apply deployment, security, governance and documentation practices.	Chapter 7; Appendix D	[Student to assess]	—

The exact CO wording should be checked against the official course handout if it differs from the template. The student and examiner columns are intentionally left for completion during submission/evaluation.

APPENDIX C – DELUGE SOURCE CODE

The complete application export is the source of truth and is supplied separately as Solar_Installation_Management_System.ds. The excerpts below transcribe the core logic visible in that file; they are included for review, not as a replacement for importing and checking the full export in Creator.

DS-01 — Auto-Assign Technician for High Priority

```
// Form: maintenance_request // Trigger: On Add -> On
Success maintenanceRequest = maintenance_request[ID
== input.ID]; if(maintenanceRequest.priority ==
"Critical") {     availableTechnicians =
technician[availability_status == "Available"];
if(availableTechnicians.count() > 0)     {
firstTechnician = availableTechnicians;
maintenanceRequest.assigned_technician =
firstTechnician.ID;     }     else     {
loggedInUserEmail = zoho.loginuserid;
loggedInTechnician = technician[email ==
loggedInUserEmail];
if(loggedInTechnician.count() > 0)         {
maintenanceRequest.assigned_technician =
loggedInTechnician.ID;          }
}
}
```

Review note: verify whether the collection-to-record assignment is valid in the target Creator context. Prefer explicit single-record selection with a deterministic sort and test all fallback paths.

DS-02 — Prevent Low Stock Submission

```
// Form: equipment_inventory // Trigger: On Add or Edit ->
On Validate if(input.quantity_available <
input.minimum_stock_level) {     alert "Error: Equipment
quantity cannot be below minimum stock level. Current: "
+ input.quantity_available.toString()
        + ", Minimum Required: "
+
input.minimum_stock_level.toStrin
g();     cancel submit; }
```

DS-03 — Notify Customer

```
// Form: customer //
Trigger: On Add -> On
Success sendmail (
    from : "zoho.adminuserid"     to : email     subject : "Record Added"
message : "Dear ${customer_name}, Welcome to SolarCare! Your customer details
have been successfully regist     content type : Plain Text     attachments :
template : _default1 as pdf )
```

The export includes longer HTML email text than the concise excerpt above. Verify the actual email template, attachment and content type in Creator before deployment.

DS-04 — Technician email

```
// Form: technician // Trigger: On Add -> On Success sendmail (     from :
"zoho.adminuserid"     to : email     subject : "Record Added"     message :
"[Current exported template contains employment-offer wording; replace with
approved message]"     content type : Plain Text     attachments : template :
_default1 as pdf )
```

The exact export contains a lengthy template referencing an employment offer at “ABC solars”. It is not reproduced in full here because it appears mismatched to the workflow purpose; inspect the source export and correct it before use. Keep the full .ds file with the submission for exact configuration provenance.

APPENDIX D – USER MANUAL

D.1 Sign in and open the application

1. Open the live Zoho Creator application URL supplied by the administrator. 2. Sign in with an authorised account. 3. Confirm that the navigation menu and Dashboard page appear. If the application is not published or shared, ask the app owner to complete access configuration.

D.2 Register a customer

1. Open Customer. 2. Enter customer name, phone, email, address and customer type. 3. Review the details and submit. 4. Confirm that the record appears in the customer report. 5. If email notification is enabled, check the controlled test inbox and verify the message and attachment.

D.3 Record a solar installation

1. Open Solar Installation. 2. Select the customer. 3. Enter installation name, site address, system capacity in kW, panel count/type, inverter model/serial, installation date, warranty duration and warranty expiry date. 4. Review the installation status and submit. 5. Confirm the record in the standard installation report.

D.4 Maintain equipment inventory

1. Open Equipment Inventory. 2. Enter equipment name/category, model/serial number, quantity available, minimum stock level, unit and storage location. 3. Ensure quantity available is not below the minimum stock level. 4. Submit and verify the record. If the rule is triggered, correct the quantity or minimum value according to the approved stock policy.

D.5 Maintain technician records

1. Open Technician. 2. Enter name, phone, email, specialisation and availability status. 3. Submit and verify the report entry. 4. Confirm the outgoing email is appropriate before using the workflow with real recipients.

D.6 Create and update a maintenance request

1. Open Maintenance Request. 2. Select the customer and installation where applicable. 3. Enter request date, issue category, description, priority and status. 4.

Submit. 5. For Critical priority, check whether a technician was assigned by the workflow. 6. During service, update diagnosis, action taken, parts used, service notes, completion date and status. 7.
Verify the report shows the updated record.

D.7 Schedule maintenance and complete checklist

1. Open Maintenance Schedule and select customer/installation. 2. Enter maintenance type, last service date, interval, next due date, schedule status, notes and assigned technician. 3. Save and verify the report. 4. After inspection, open Maintenance Checklist, select the checklist item and condition, enter observations and corrective action, link the maintenance request, and save.

D.8 Use reports and dashboard

Use the standard reports to review full record lists and grouped reports to inspect records by issue category, installation status, maintenance type, technician specialisation, equipment category, customer type and checklist item. On the Dashboard, compare KPI values with known test records and inspect chart groupings. If values appear inconsistent, verify the source records and report criteria.

D.9 Operational cautions

Do not enter real customer data into a test environment unless approved. Do not treat the configured assignment or email workflows as verified until tested. Review the technician email wording, confirm the timezone, and ensure that access permissions match each user's job responsibilities.

APPENDIX E – EVALUATION RUBRIC

S.No.	Criterion	Indicators	Max. marks	Marks awarded
1	Problem definition and requirement analysis	Clarity of problem, measurable objectives, complete FR/NFR, use cases	10	
2	System design	Architecture, ER/data model, form specifications, workflow design, permission matrix	10	
3	Form and report implementation	Variety of field types, lookups/subforms, report types, UI Builder dashboards	15	
4	Deluge Scripting & Workflows	Correctness and depth of scripts; form, schedule, approval, Blueprint, batch workflows	15	
5	Integration, RBAC & AI	Zoho/external integrations, connections, roles & permissions, Zia/analytics	10	
6	Testing, Deployment, Security & Governance	Test coverage, environments, optimisation, security controls, audit	10	
7	Live Demonstration	End-to-end working of the application across roles; feature checklist verified	10	
8	Project Report	Adherence to template and format, clarity, figures, references	10	
	Viva-voce	Understanding of the platform, design decisions and low-code concepts	10	
	Total		**100**	

Performance descriptors

Level	% of max marks	Descriptor
Excellent	90–100	Feature is implemented completely and correctly; student explains design choices and evidence confidently.
Good	70–89	Feature is implemented correctly with minor gaps in edge cases or documentation.
Satisfactory	50–69	Core behavior exists but validation, usability, evidence or test coverage is incomplete.
Needs improvement	Below 50	Feature is incomplete, unreliable, undocumented or not demonstrated.

Signature of Internal Examiner

Name: _____

Signature of External Examiner

Name: _____

APPENDIX F – VIVA-VOCE RECORD SHEET

(For Examiner Use Only)

Student Name	SELVAGANAPATHY T	Register No.	241501194
Department / Section	AIML	Date of Viva	
Application URL Verified	Yes / No	Demo Duration	_____min

Table F.1: Viva-Voce Questions and Responses

Q.No.	Question / Topic Asked	Unit	Response (E / G / S / NI)	Remarks
1				
2				
3				
4				
5				
6				
7				
8				

E – Excellent G – Good S – Satisfactory NI – Needs Improvement

Table F.2: Marks Summary

Component	Max. Marks	Marks Awarded
Criteria 1–8 (Appendix E)	90	
Viva-Voce	10	
TOTAL	**100**	

Examiners' Remarks:

_____

_____

INTERNAL EXAMINER **EXTERNAL EXAMINER**

Name: _____ Name: _____